

Software Test Plan (STP) - Task Manager Application

Project: Task Manager Application

Version: 1.0

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Date: 02-10-2026

Status: Draft

1. Introduction

Purpose: This document defines the test plan for the Task Manager Application v1.0. It outlines the testing objectives, scope, strategy, resources, schedule, responsibilities, security validation, and requirement traceability.

Scope: Testing covers user registration, authentication, task creation, viewing, editing, deletion, assignment, status and priority management, search/filtering, and dashboard progress tracking. Testing also covers applicable performance, security, usability, reliability, maintainability, and testability requirements.

References: Task Manager Application SRS v1.0, Task Manager Application Software Architecture and Design Specification v1.0, Software Engineering Mini-Project Deliverables Part-1, Requirements Traceability Matrix (RTM).

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design Specification), STP (Software Test Plan), RTM (Requirements Traceability Matrix), API (Application Programming Interface), JWT (JSON Web Token), REST (Representational State Transfer), FR (Functional Requirement), NFR (Non-Functional Requirement), SR (Security Requirement).

2. Test Items

- User registration and authentication module
- Task management module
- Task assignment, status, and priority module
- Search and filtering functionality
- Dashboard and progress tracking
- Node.js/Express REST API
- Database persistence and access control

3. Features to be Tested

Features mapped to SRS requirement IDs:

- FR-01: User registration with unique email and password
- FR-02: User authentication before protected functionality
- FR-03: Task creation with title, description, priority, and deadline
- FR-04 to FR-06: Task viewing, editing, and deletion
- FR-07 to FR-09: Task assignment, status updates, and priority management
- FR-10: Search and filtering of authorized tasks
- FR-11: Dashboard-based task progress tracking
- NFR-01: Normal task-management requests within 3 seconds under normal conditions
- NFR-02: Protection against unauthorized access to protected functions
- NFR-03 to NFR-06: Interface consistency, data preservation, maintainability, and independent testability

4. Features Not to be Tested

- Features outside the approved SRS scope, such as notifications or advanced collaboration features
- Production infrastructure administration and cloud-provider internals
- Third-party services not included in the implemented application
- Database engine internals beyond the application-level persistence and access behavior

5. Test Approach / Strategy

Levels:

- Unit tests (module-level frontend/backend components)
- Integration tests (React frontend ↔ Node.js/Express API ↔ database)
- System tests (end-to-end Task Manager functionality)
- Acceptance tests (UAT against approved SRS requirements)

Types:

- Functional testing (registration, authentication, task operations, assignment, status, priority, search/filtering, dashboard)
- Regression testing after changes to major modules
- Performance testing (normal request response time)
- Security testing (authentication, authorization, input validation, sensitive-data handling)
- Usability testing (navigation, forms, validation and error messages)

Entry Criteria: Stable build delivered, test data available, frontend/backend/database test environment ready, and required requirements/design documents available.

Exit Criteria: 100% of planned test cases executed, no critical defects open, all security validation checks completed, and applicable acceptance criteria satisfied.

5.1 Security Validation

- Verify authentication is required before protected task-management operations (SR-01).
- Verify authorization checks prevent access to or modification of tasks without permission (SR-02).
- Verify passwords and authentication credentials are not exposed in API responses or normal application logs (SR-03).
- Test server-side validation for invalid, missing, or unexpected user/task input (SR-04).
- Verify protected API communication uses HTTPS when deployed in a secure environment.
- Test authentication and authorization failure responses for appropriate 401/403 behavior.

6. Test Environment

Hardware: Development/test computers capable of running the web browser, React.js frontend, Node.js/Express backend, and database.

Software: React.js frontend, Node.js with Express.js backend, configured database, supported modern web browser, and the Task Manager Application test build.

Tools: Browser developer tools, API testing tools such as Postman or equivalent, Git, and project test tools selected during implementation.

Test Data: Dummy user accounts, valid and invalid credentials, sample tasks with different priorities/statuses/deadlines, and task-assignment data.

7. Test Schedule

Milestones:

- Test plan preparation: 02-Oct-2026
- Test case design and traceability: 02-Oct-2026
- Test environment setup: After stable implementation build is available
- Test execution start: After environment readiness
- Test execution end: After all planned test cases are completed
- UAT and final test summary: After system testing and defect closure

8. Test Deliverables

- Test Plan (this document)
- Test Cases and test data
- Test Scripts, where automation is used
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix updates
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility

QA Lead	Team 4	Prepare plan, coordinate test execution, and maintain test status.
Test Engineer	Team 4	Design and execute test cases, record results, and log defects.
Developer	Team 4	Support defect analysis, fixes, and regression testing.
Product Owner	Team 4	Review test results and approve readiness for the next stage.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Prepare test cases and test data early and request stable builds before execution.
Test environment or database downtime	Maintain a reproducible local test setup and restore required services before resuming tests.
Defects in authentication or authorization	Prioritize security validation, isolate blocking defects, and perform regression testing after fixes.

11. Assumptions & Dependencies

- React.js, Node.js/Express.js, and database test environment will be available before execution.
- Test data for registered users and tasks will be prepared before execution.
- The implemented APIs and UI will follow the requirements and architecture defined in the SRS and SAD.
- Team members will provide stable builds and access required for testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for an extended period or prevents planned execution.
- A blocking defect prevents execution of a substantial portion of the planned test cases.
- The application build is too unstable to produce meaningful test results.

Resume testing if:

- Blocking defects are resolved or an approved stable build is provided.
- The test environment is restored and required services are available.

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. Planned test cases:

- TC-01 (FR-01): Register a new user with a unique email and valid password; verify successful account creation.
- TC-02 (FR-02, SR-01): Log in with valid credentials and verify access to protected functionality; verify unauthenticated access is rejected.
- TC-03 (FR-03, SR-04): Create a task with valid title, description, priority, and deadline; verify the task is stored.
- TC-04 (FR-04): View tasks and verify only authorized task information is displayed.
- TC-05 (FR-05): Edit an existing authorized task and verify the updated details are stored.
- TC-06 (FR-06): Delete an authorized task and verify it is no longer available.
- TC-07 (FR-07, FR-08, FR-09): Assign a task, change its status, and modify its priority; verify each change is reflected.
- TC-08 (FR-10): Search and filter tasks using supported task attributes; verify matching authorized tasks are returned.
- TC-09 (FR-11, NFR-03): Open the dashboard and verify task progress is summarized clearly and consistently.
- TC-10 (NFR-01, NFR-02, SR-02, SR-03, SR-04): Verify normal API requests meet the 3-second target under normal conditions and verify unauthorized requests, sensitive-data exposure, and invalid input are handled securely.

Each test case shall record actual result, pass/fail status, defect reference where applicable, and the corresponding SRS requirement ID.

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % test cases passed/failed
- Requirement coverage
- Defect count and severity
- Defect aging
- Security validation completion
- Performance response-time results

Reports:

- Test execution status
- Defect status report
- Requirements/test traceability status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Team 4	
Dev Lead	Team 4	
Product Owner	Team 4	